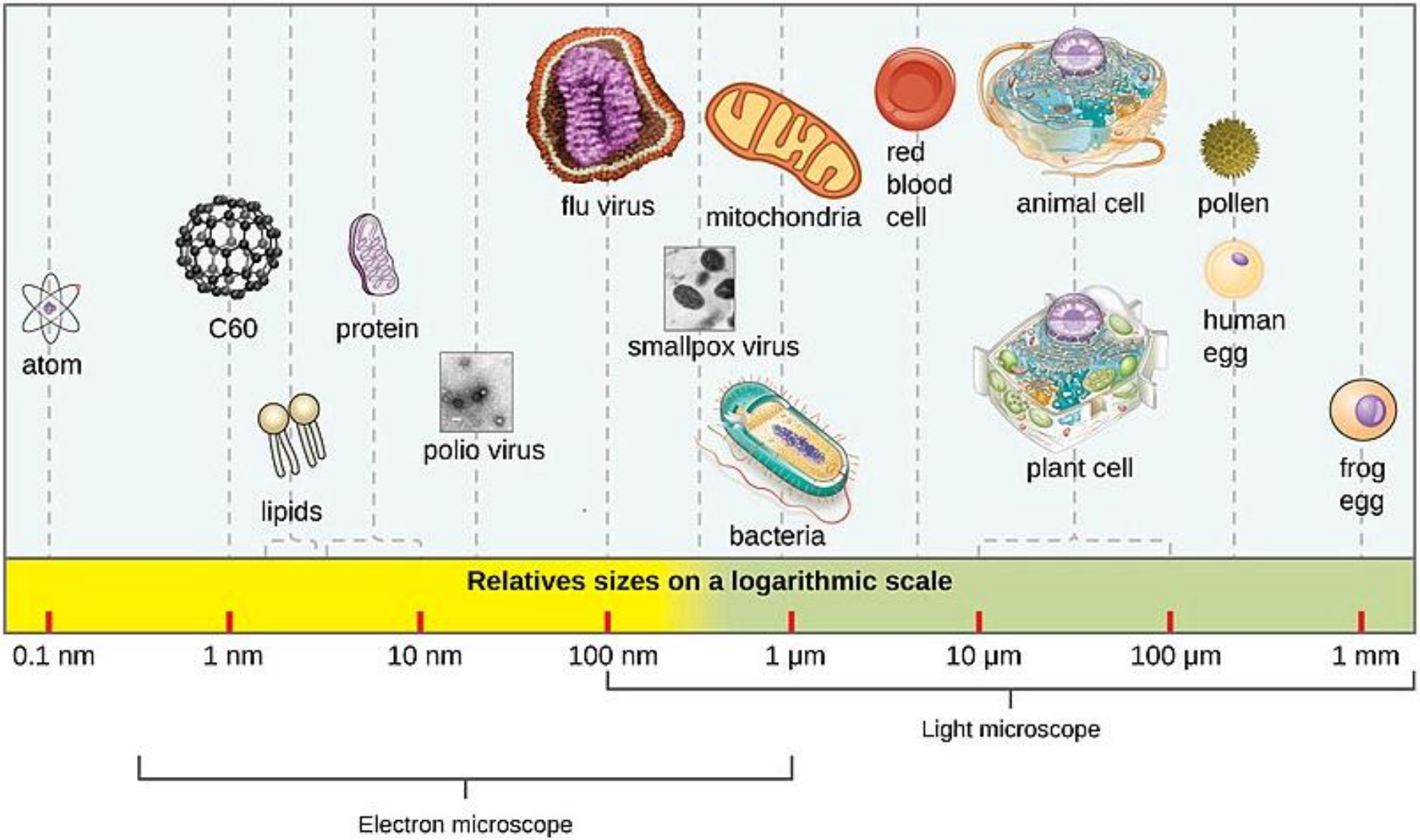
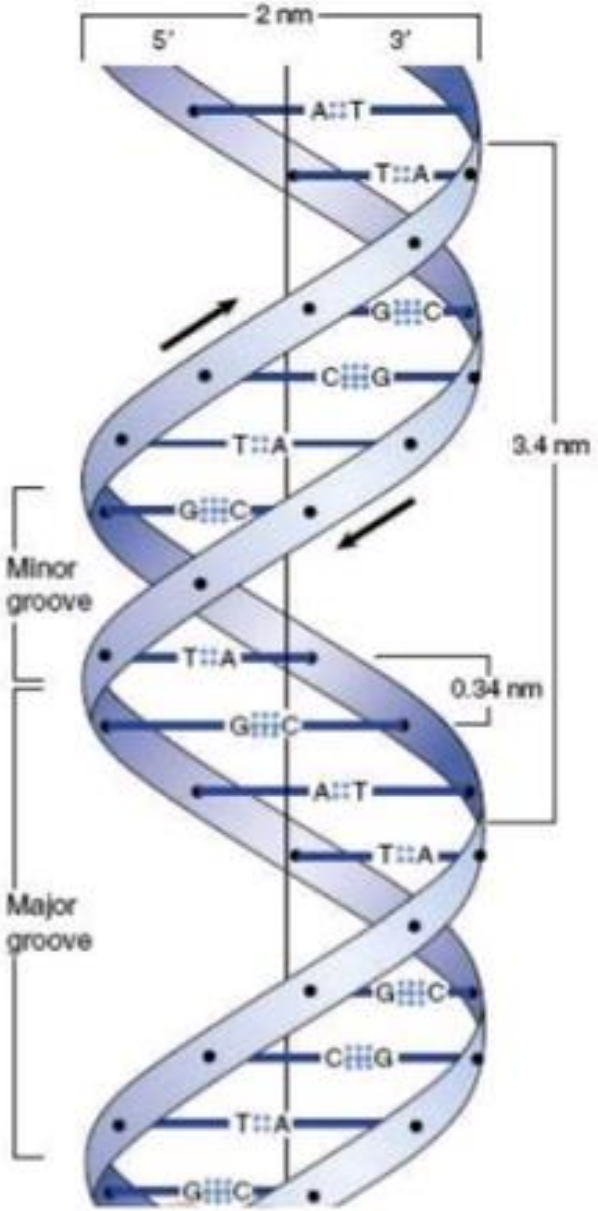
DNA, Genes, Chromosomes and Chromosomal Aberration


atom
C60
protein
lipids
polio virus
flu virus
mitochondria
smallpox virus
bacteria
red blood cell
animal cell
plant cell
pollen
human egg
frog egg
Relatives sizes on a logarithmic scale
0.1 nm
1 nm
10 nm
100 nm
1 µm
10 µm
100 µm
1 mm
Light microscope
Electron microscope

The double-helical structure of DNA

- Right-handed helix;
- The diameter of the double helix: 2 nm
- The distance between two base pairs: 0.34 nm;
- Each turn of the helix involves 10 bases pairs, 3.4 nm.



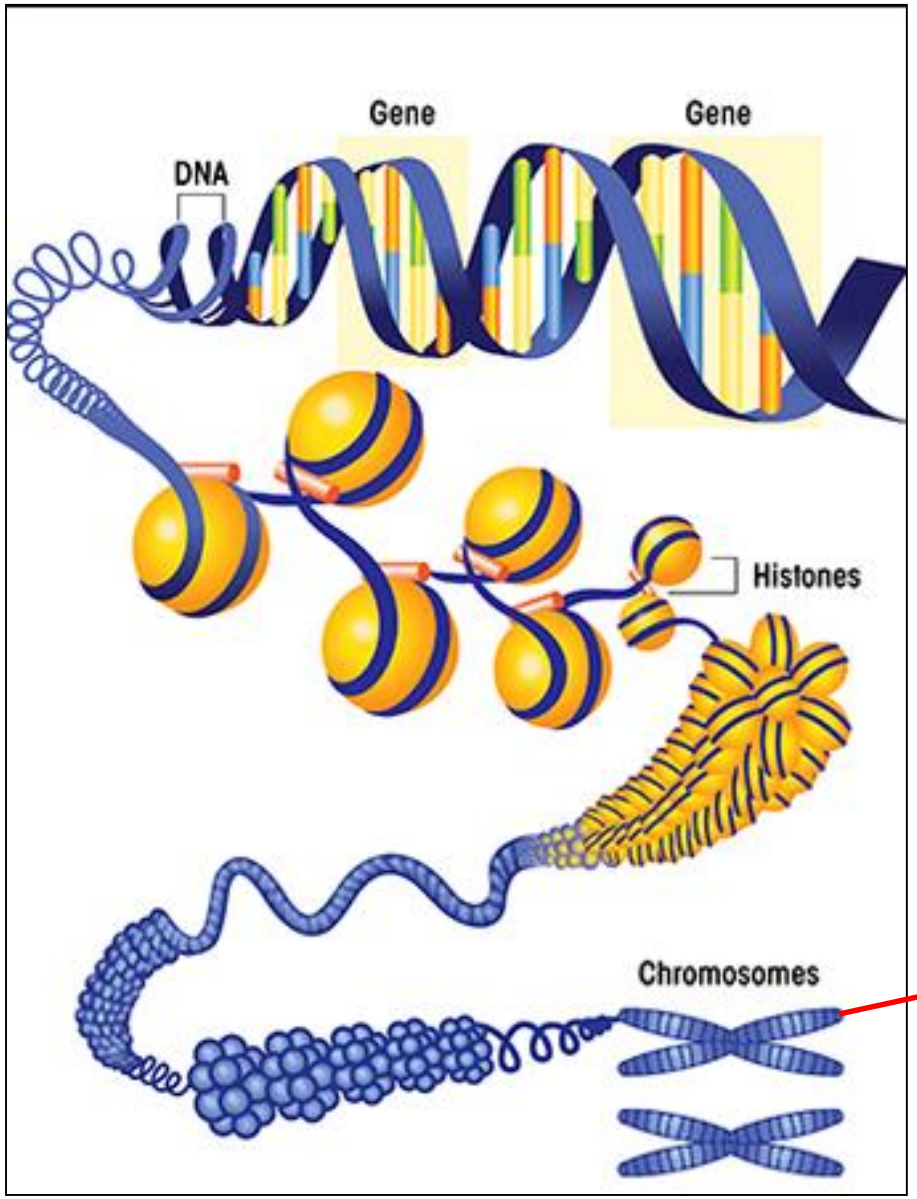
Each of us has enough DNA to reach from here to the sun and back, more than 300 times.

The haploid human genome contains approximately 3 billion base pairs of DNA packaged into 23 chromosomes. Of course, most cells in the body (except for female ova and male sperm) are diploid, with 23 pairs of chromosomes. That makes a total of 6 billion base pairs of DNA per cell. Because each base pair is around 0.34 nanometers long (a nanometer is one-billionth of a meter), each diploid cell therefore contains about 2 meters of DNA [$(0.34 \times 10^{-9}) \times (6 \times 10^{9})$].

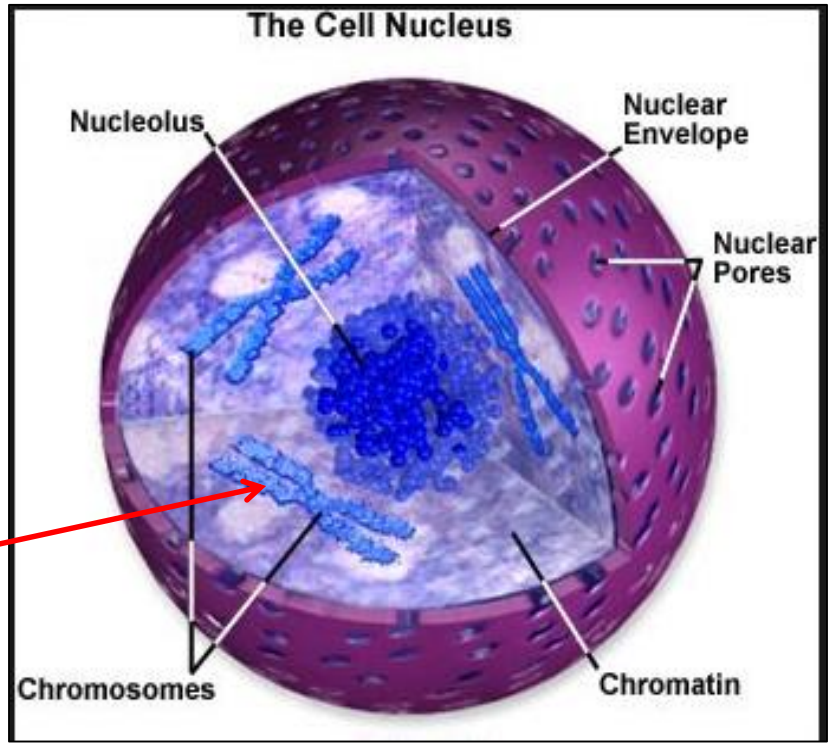
It is estimated that the human body contains about 50 trillion cells—which works out to 100 trillion meters of DNA per human. Now, consider the fact that the Sun is 150 billion meters from Earth. This means that each of us has enough DNA to go from here to the Sun and back more than 300 times, or around Earth's equator 2.5 million times!

How is this possible?

How is all of that DNA packaged so tightly into chromosomes and squeezed into a tiny nucleus?

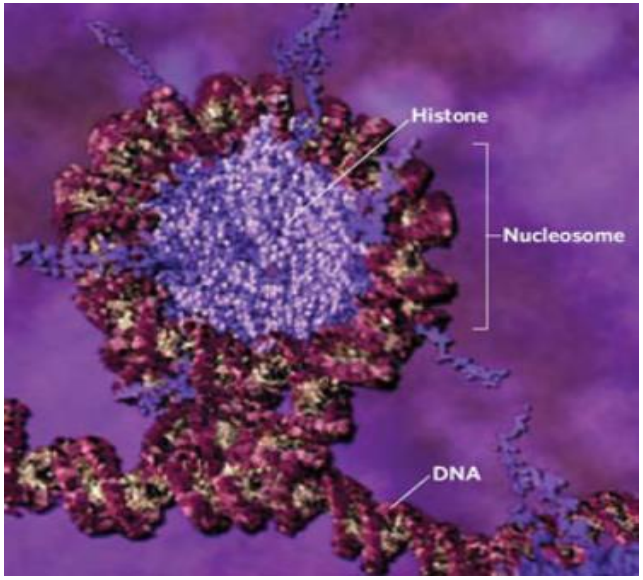


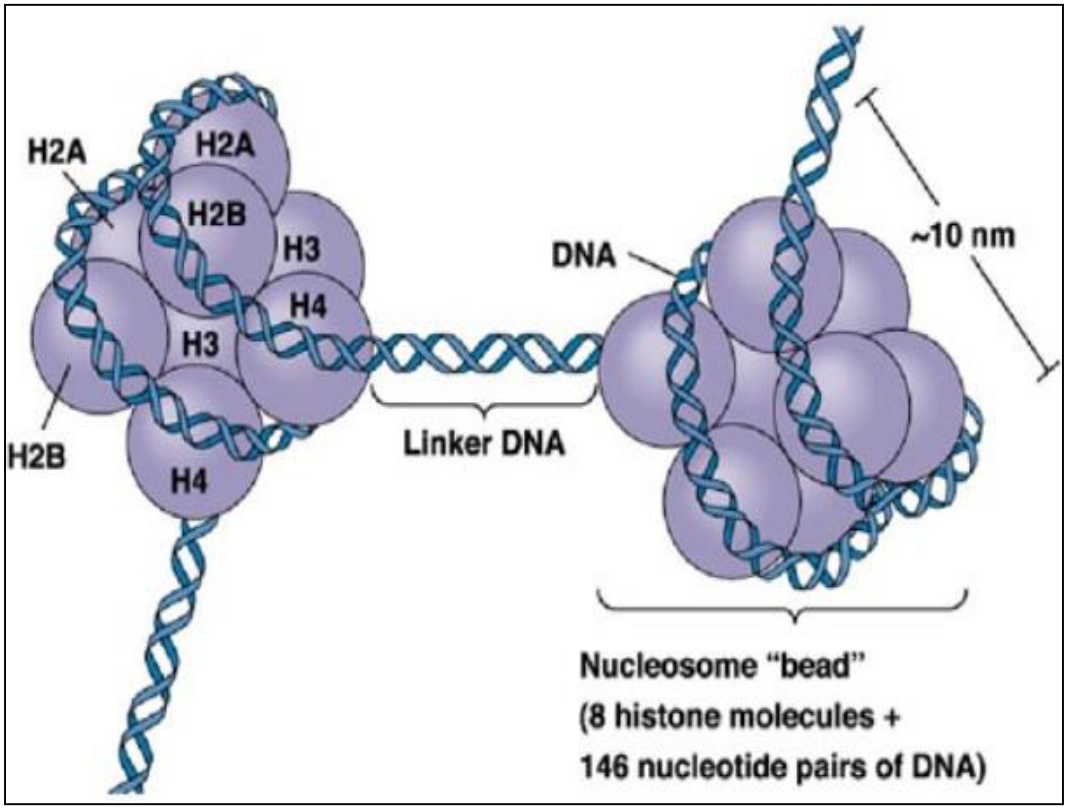
Human cells have 23 pairs of chromosomes. Each one is made up of a DNA molecule that is made up of a series of nucleotides arranged in 2 strands that spiral to form a double helix.



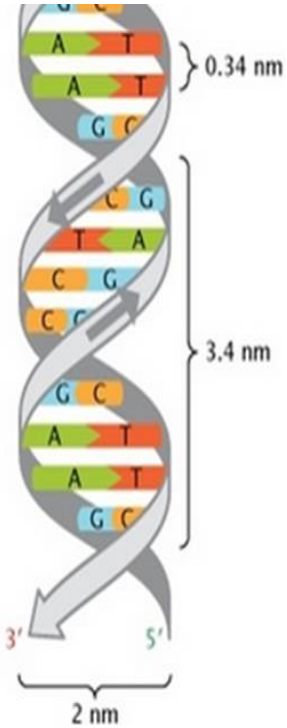
Chromosomal DNA is packaged inside microscopic nuclei with the help of Histones

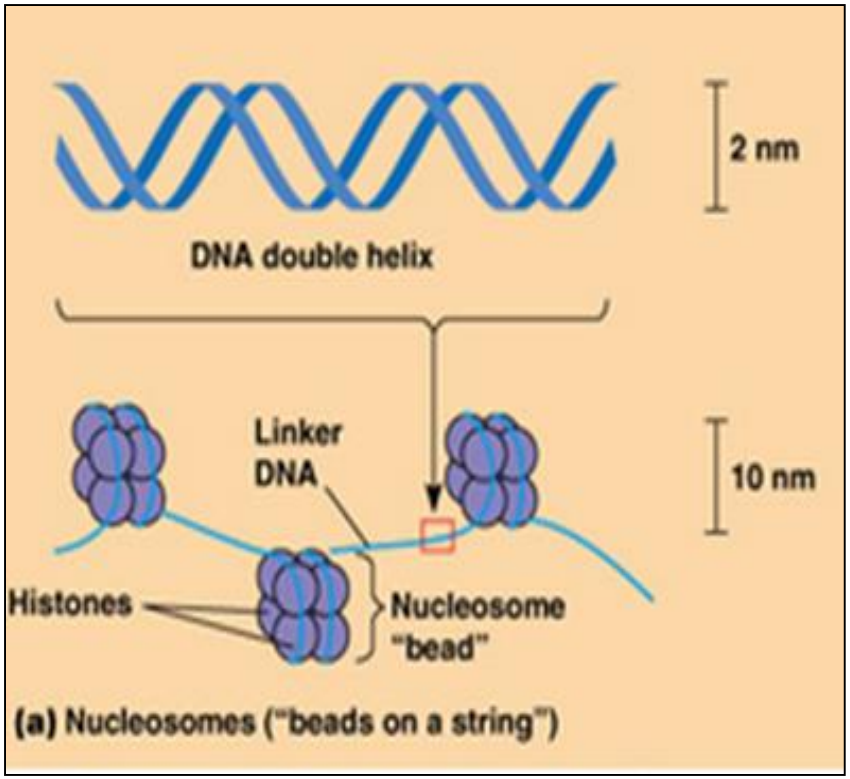
Histones are positively-charged proteins that strongly adhere to negatively-charged DNA and form complexes called nucleosomes.



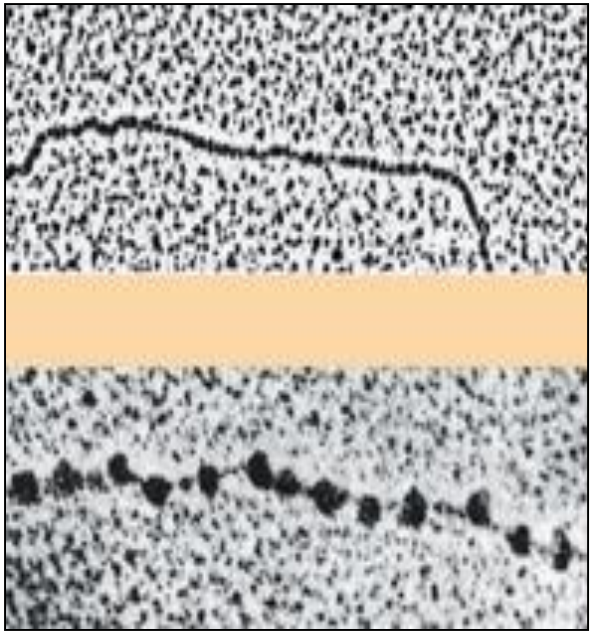


DNA Double helix to The Nucleosome



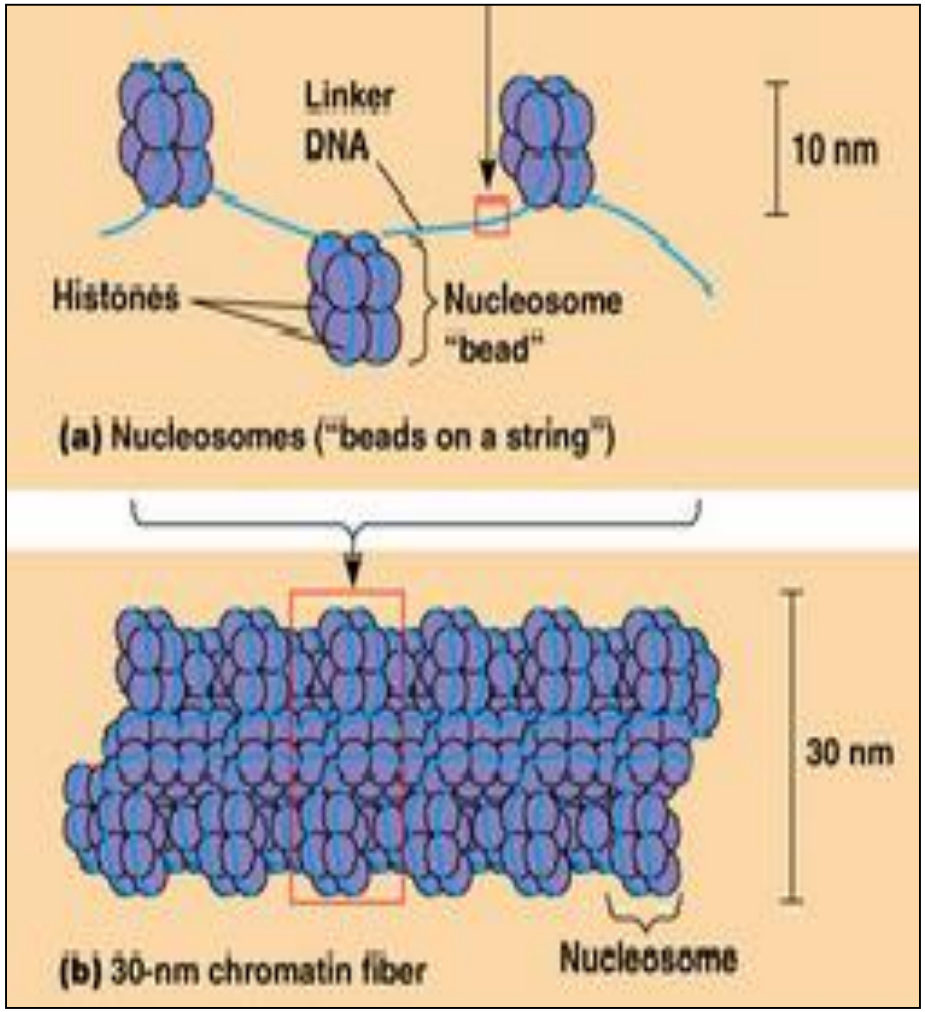


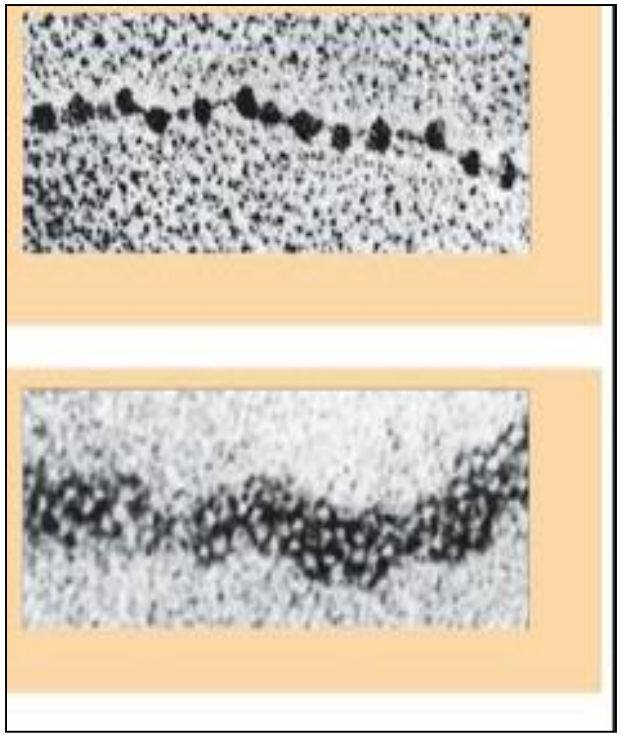
Schematic diagram of chromatin: the beads on a string

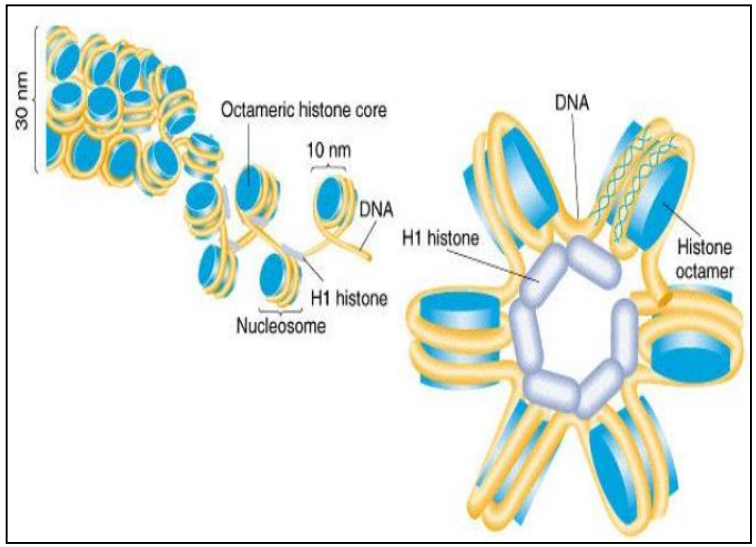


Electron micrograph of chromatin: the beads on a string

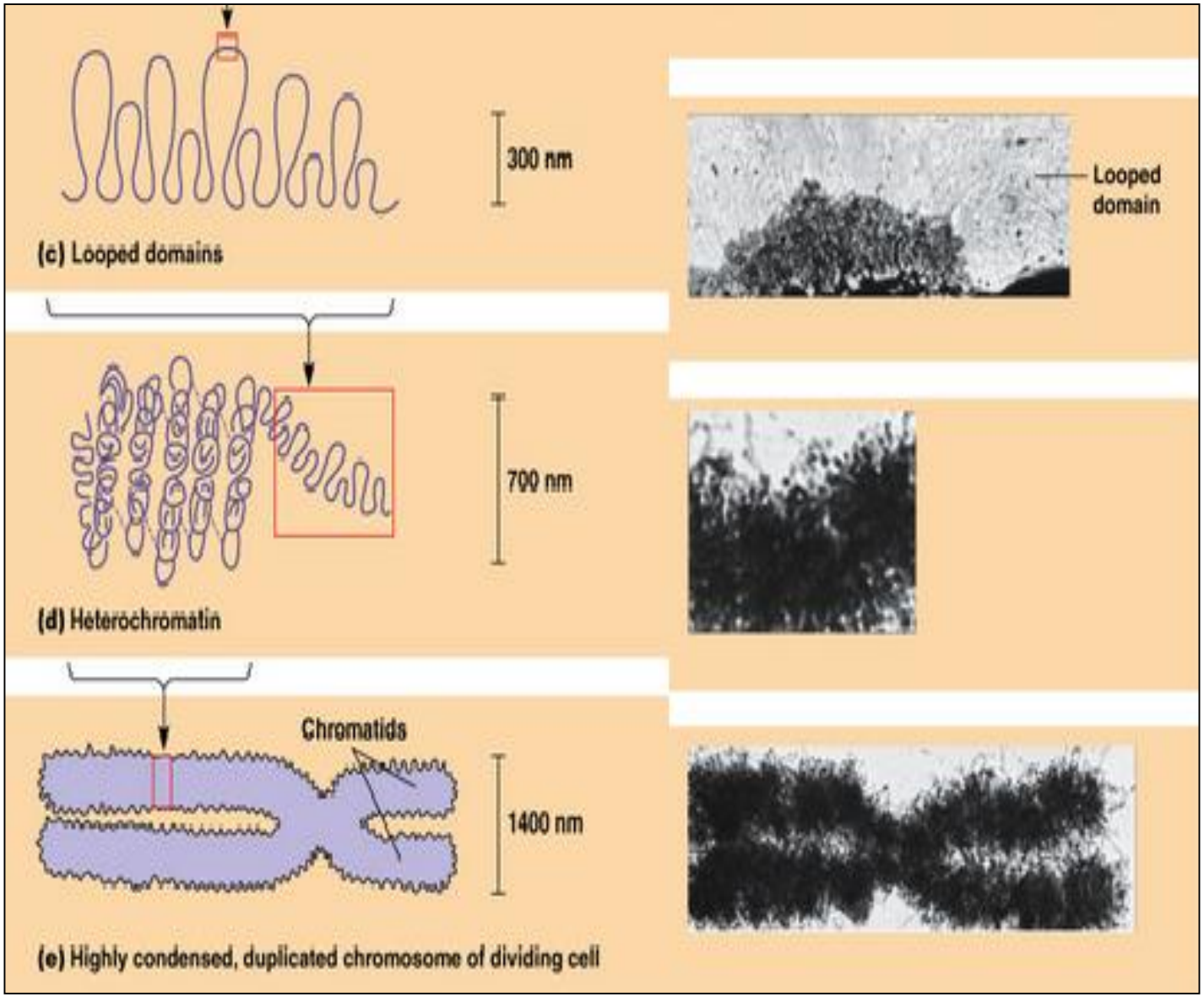
Nucleosomes fold up to form a 30-nanometer chromatin fiber







30-nanometer chromatin fiber forms loops averaging 300 nanometers in length. The 300 nm fibers are compressed and folded to wide fiber, which is tightly coiled into the chromatid of a chromosome.

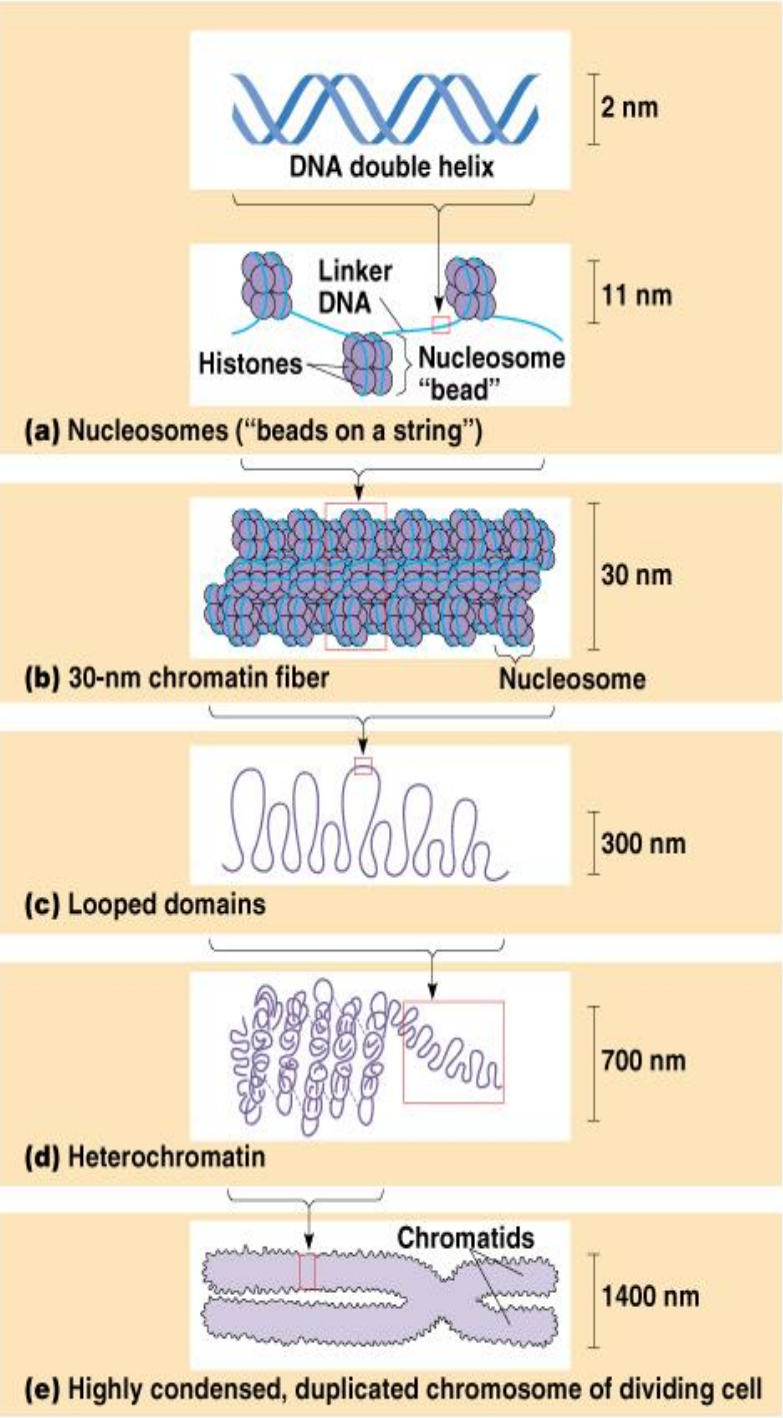


Summary

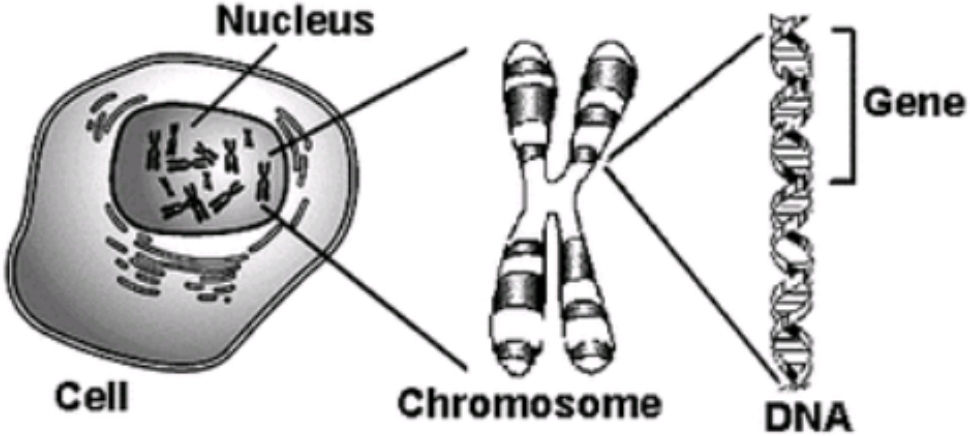
DNA molecules are packaged in the cell as structures called <u>chromosomes</u>.

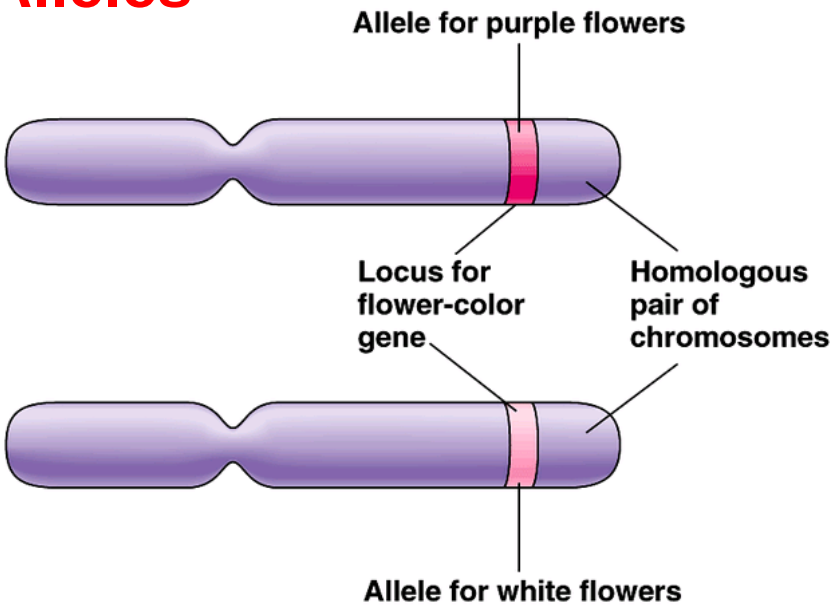
Bacteria have a single chromosome. Eukaryotes have multiple chromosomes. Humans have 46 chromosomes.

All of an organism's chromosomes make up the <u>genome</u>.



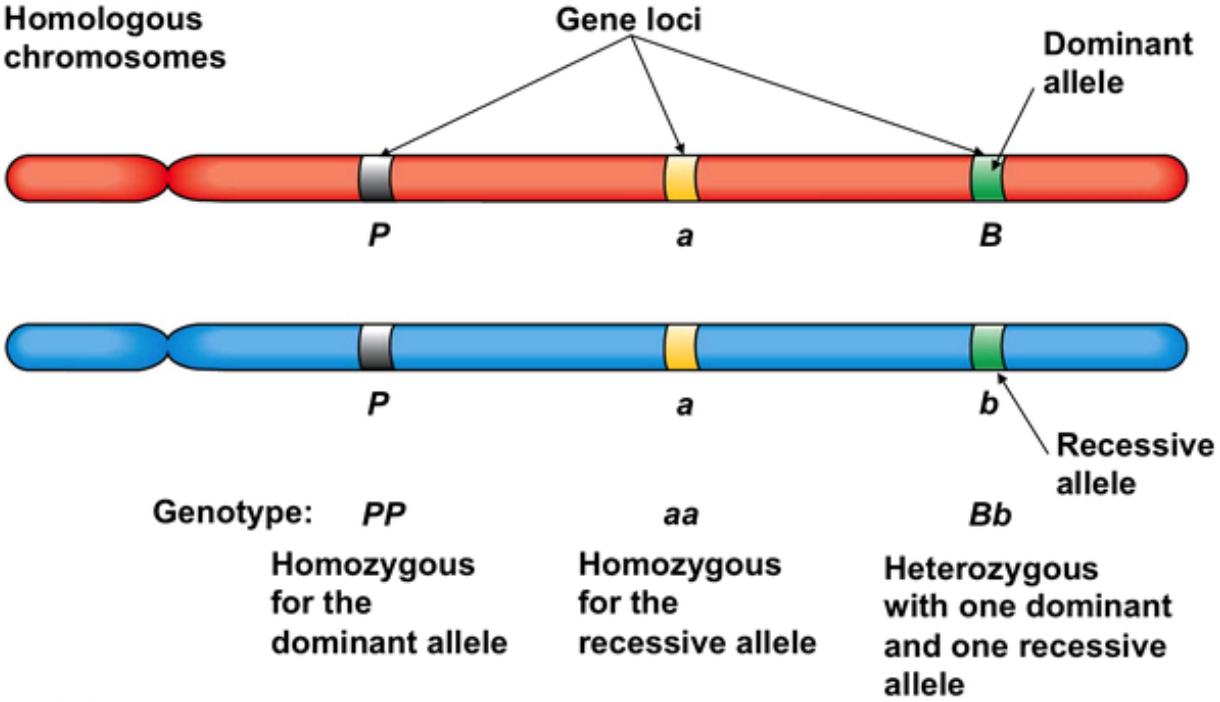
Genes and Alleles



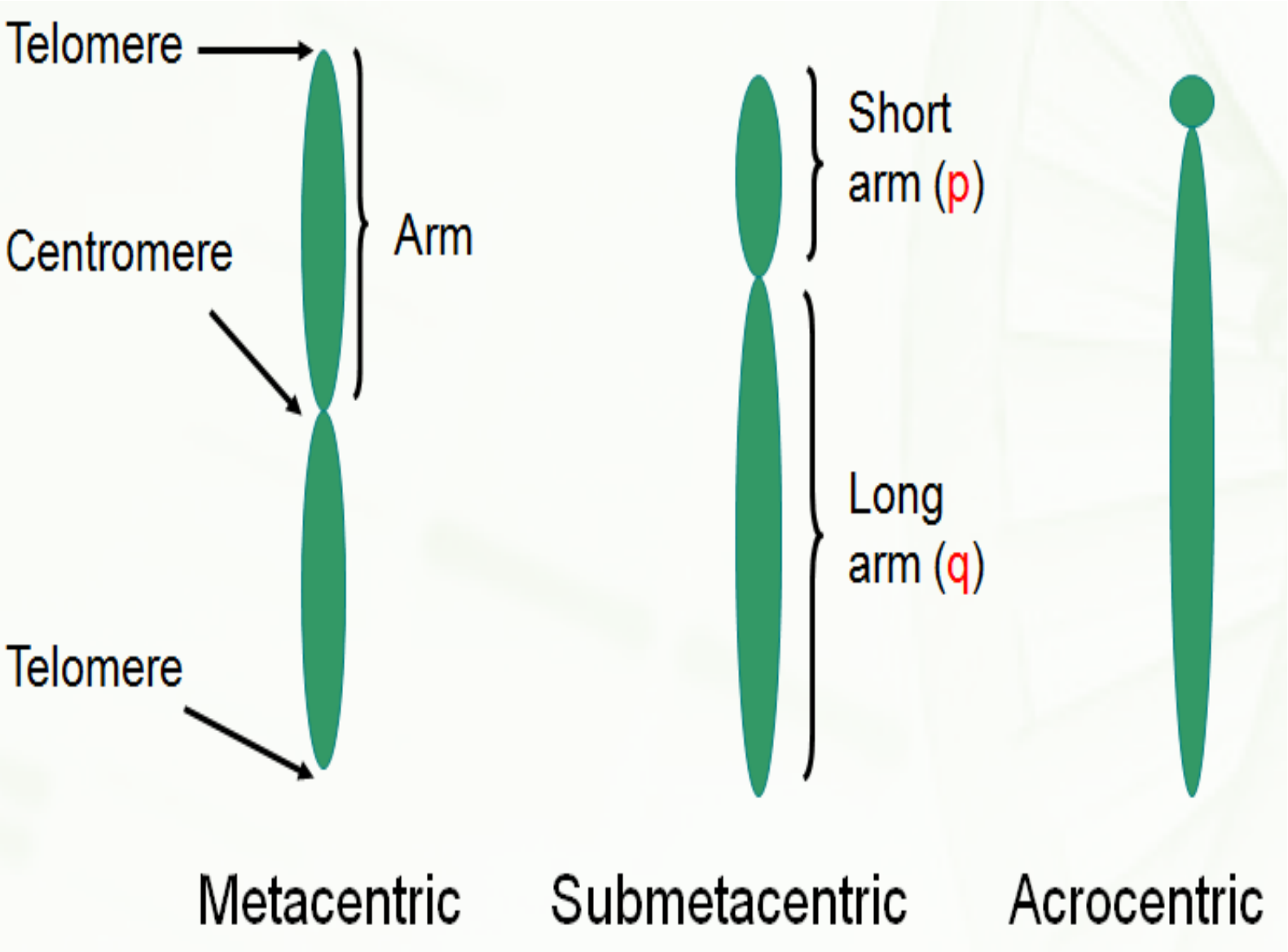


gene – segment of DNA containing instructions for a trait

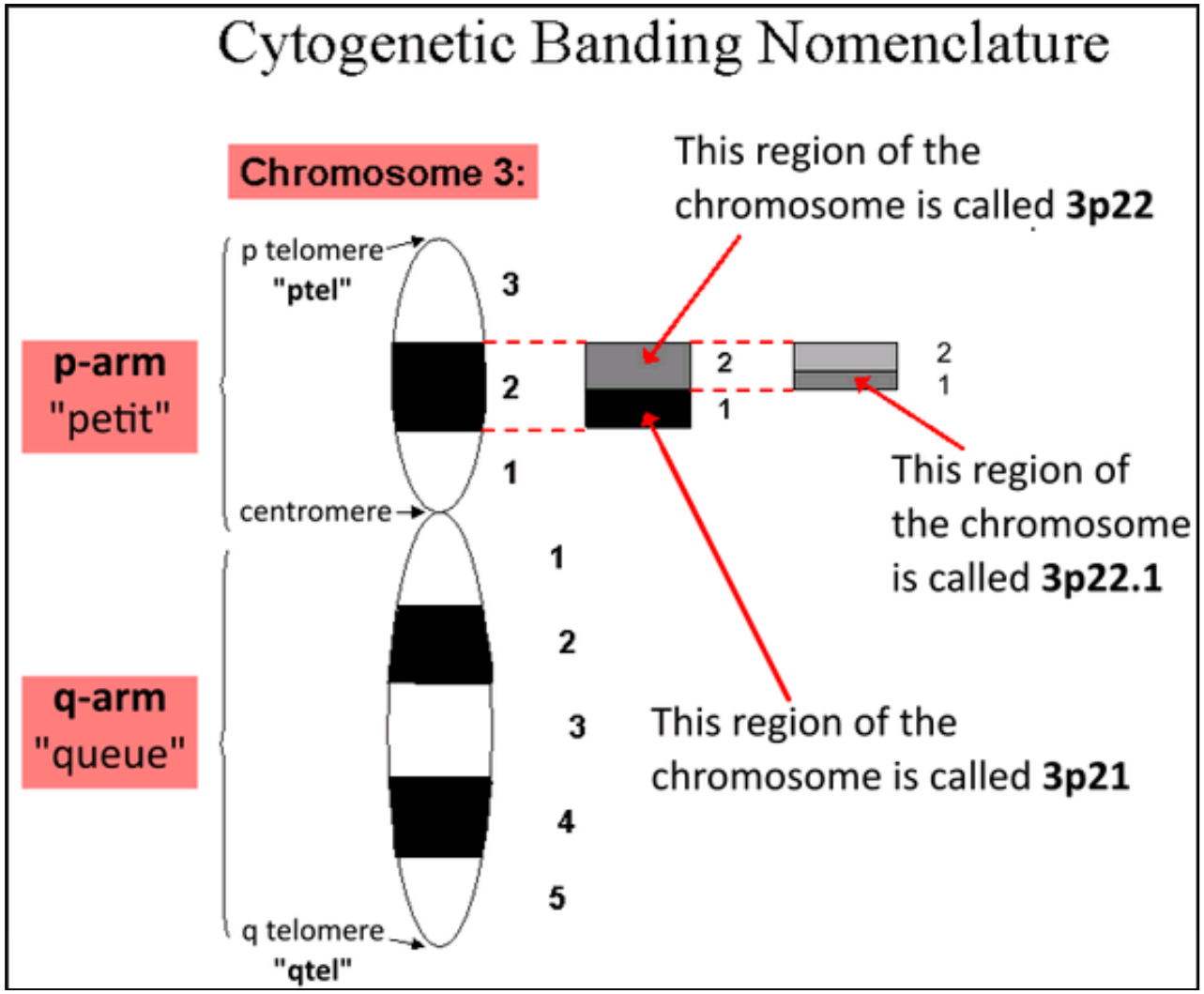
allele – one version of a gene that governs a characteristic (such as fur color)



Chromosome Morphology

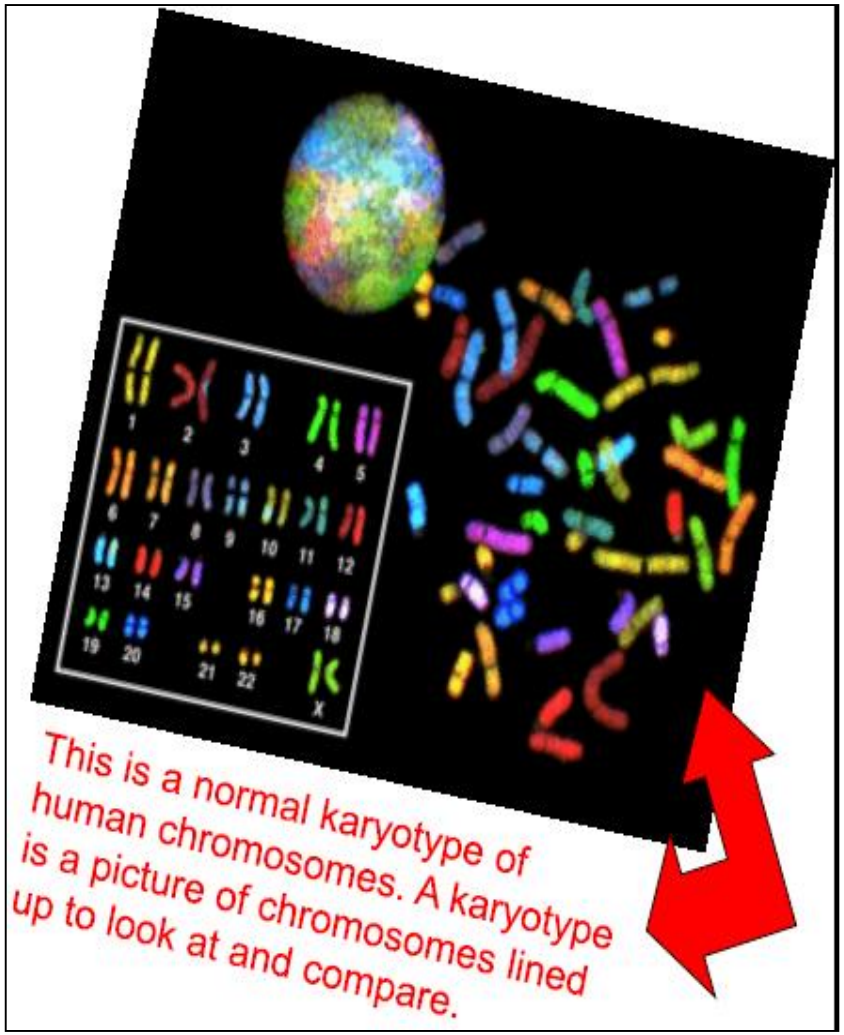


Defining Chromosomal Location

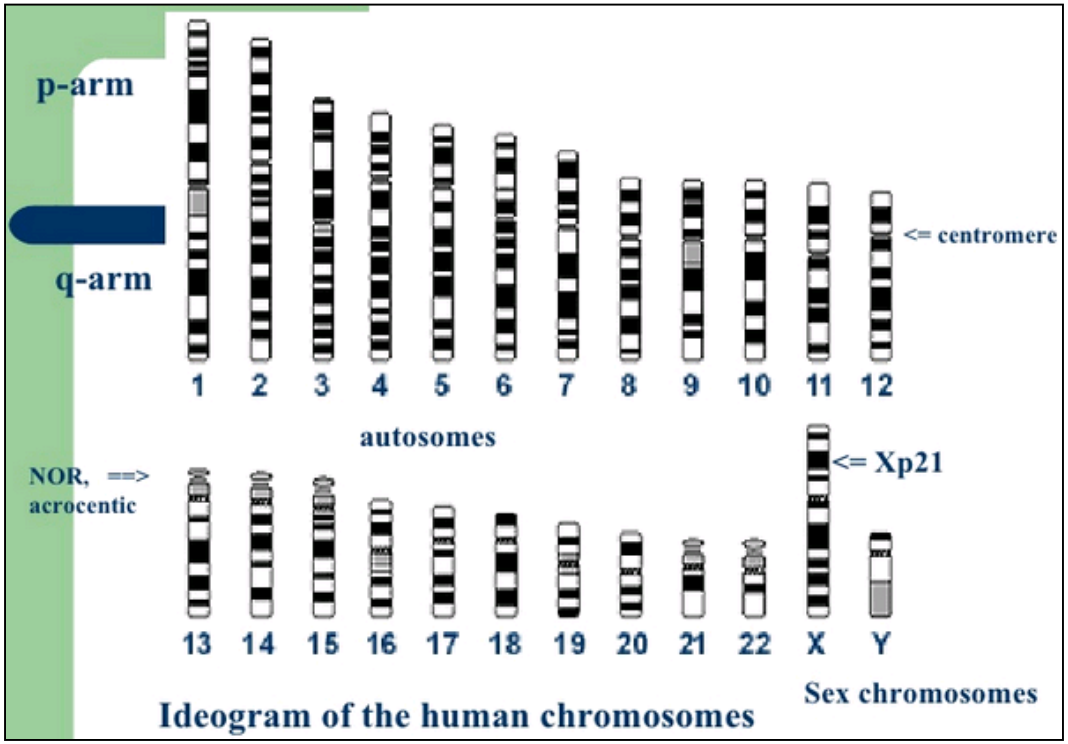


The bands are visible under a MICROSCOPE when chromosome is suitably stained. Each of the bands is numbered, beginning with 1 for the band nearest the centromere. Sub-bands and sub-sub-bands are visible at higher resolution.

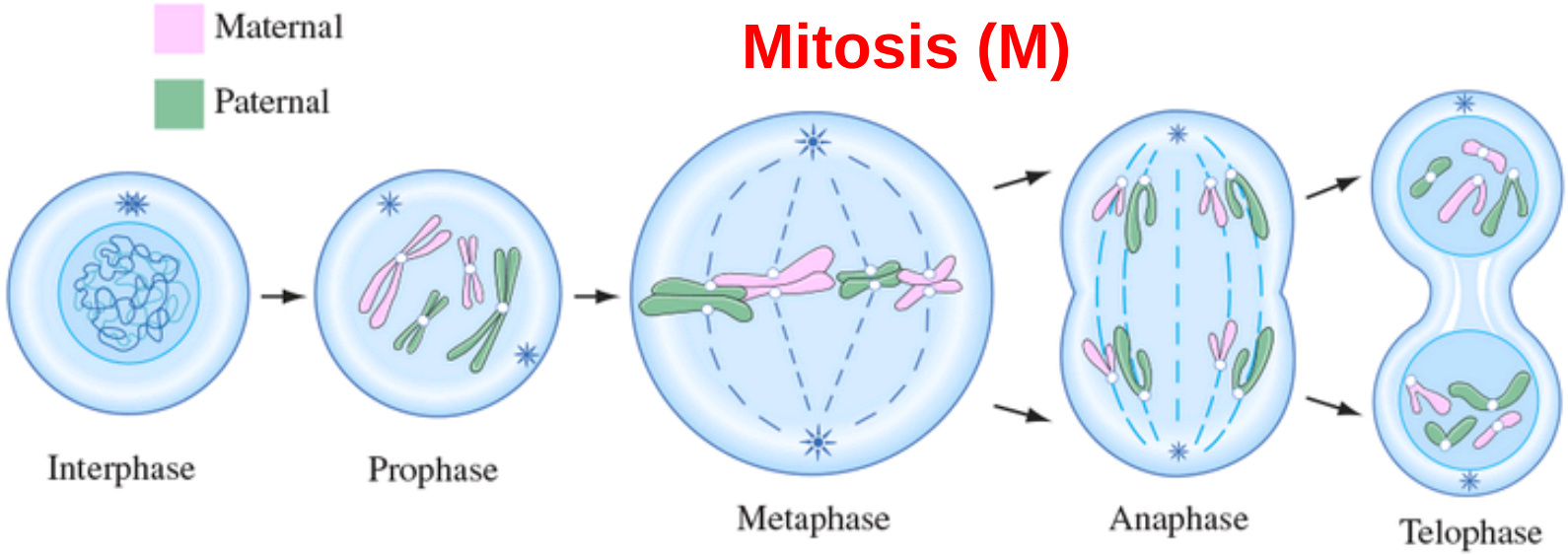
Human Genome: karyotype and idiogram

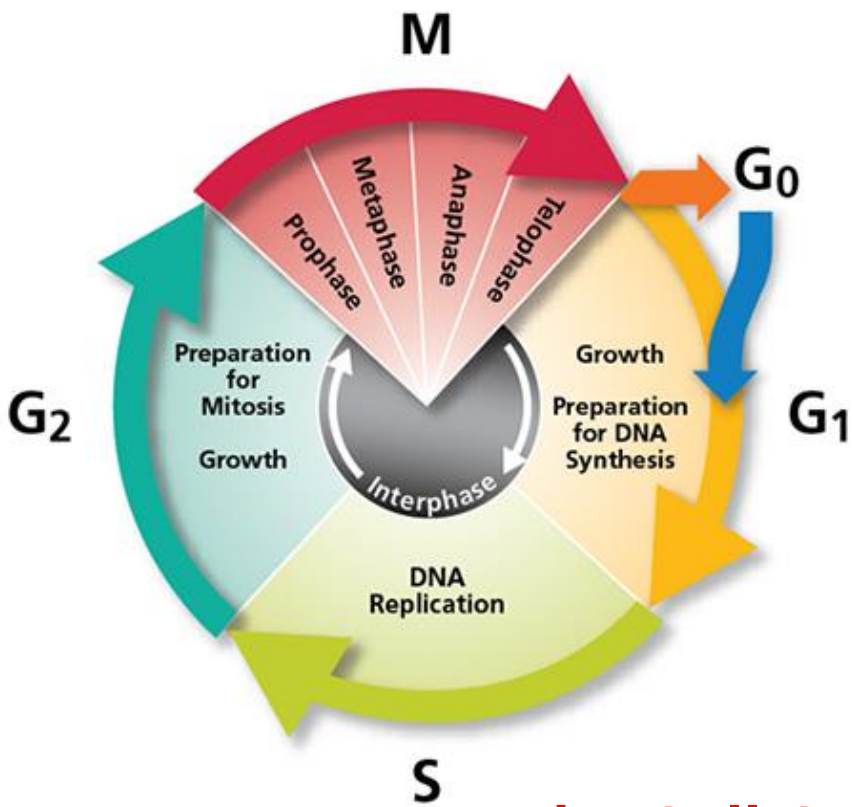


A diagrammatic representation of chromosome morphology characteristic of a species or a population is known as Idiogram



Mitosis (M)





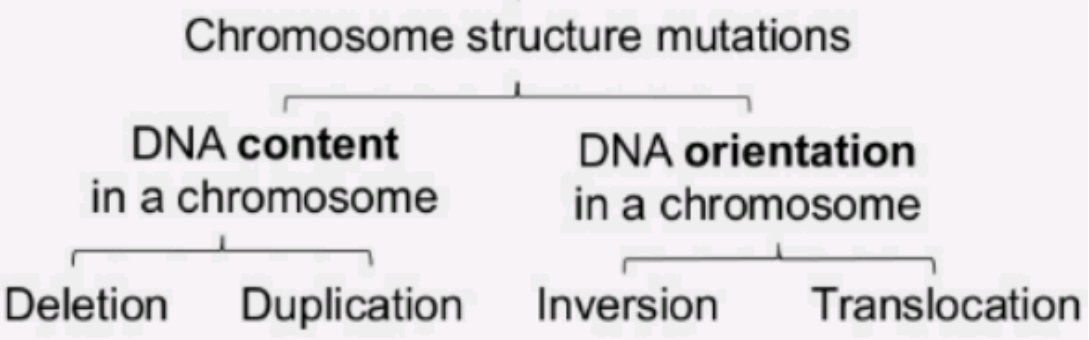
The Cell Cycle

Chromosomal Mutations

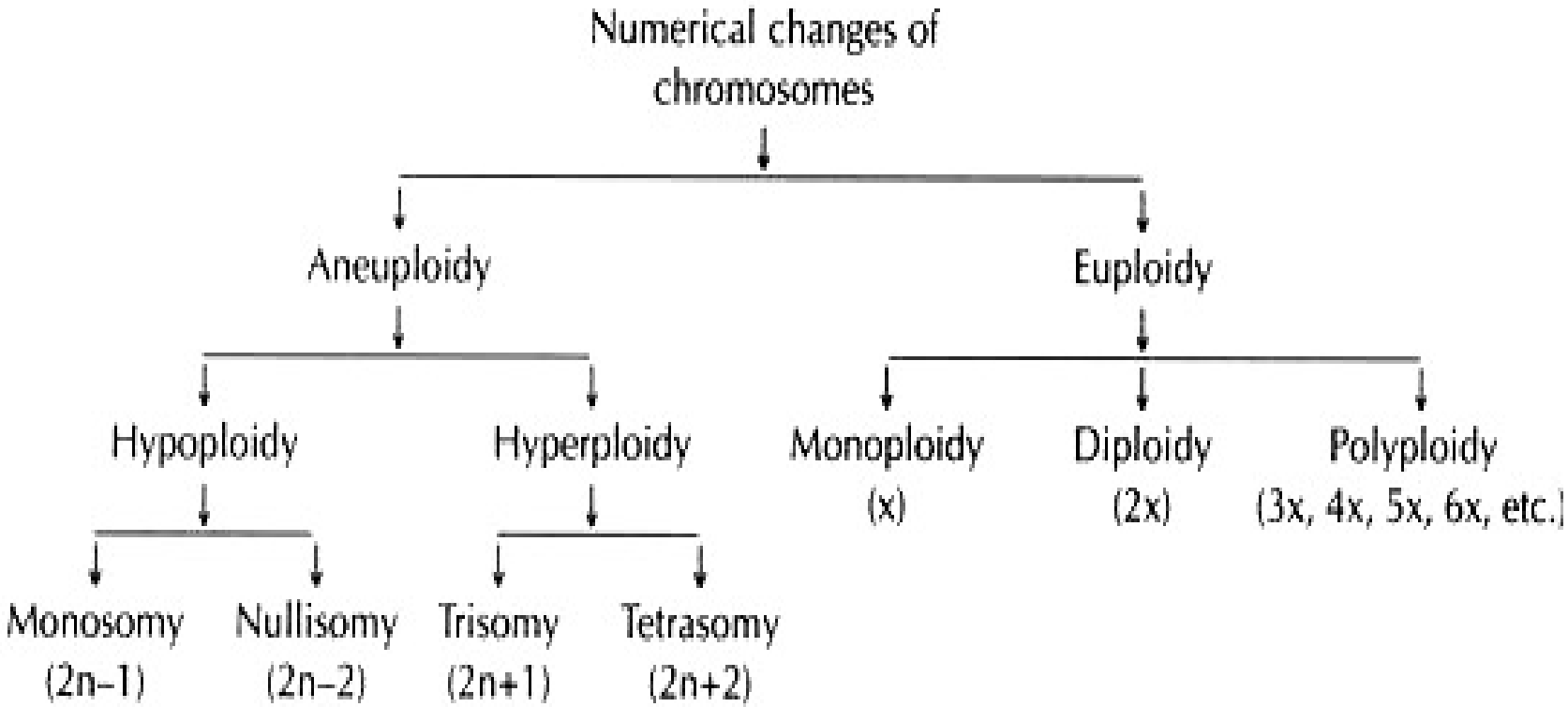
- Any change in the **structure** or **number** of chromosomes
- Large scale: Affect *many* genes

Chromosomal Mutation

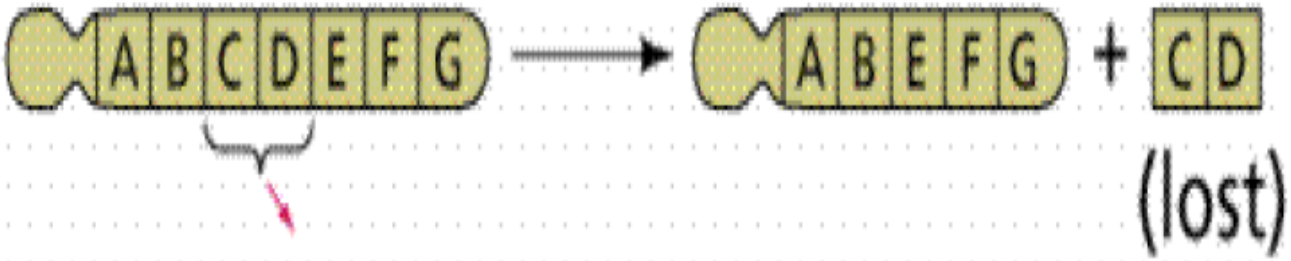
1. Chromosomal Mutation due to change in structure



2. Chromosomal Mutation due to change in number



Chromosomal Deletion



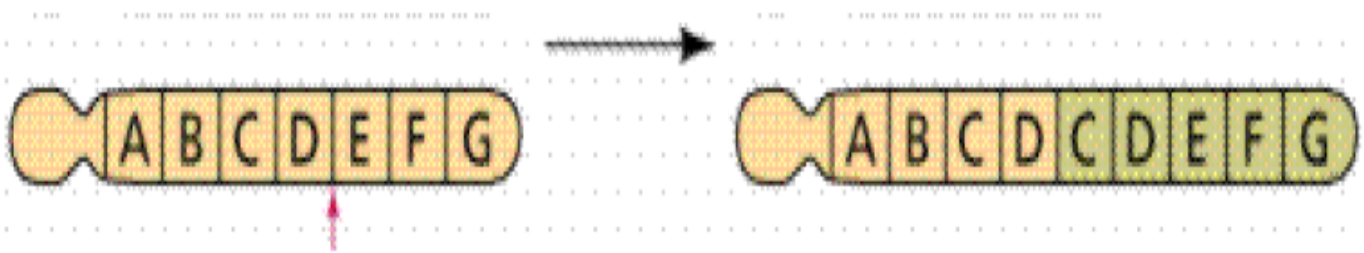
One or more genes are removed

Causes:

Wolf-Hirschhorn syndrome (severe mental retardation)

cri du chat syndrome (mewing sounds, mental retardation)

Chromosomal Duplication



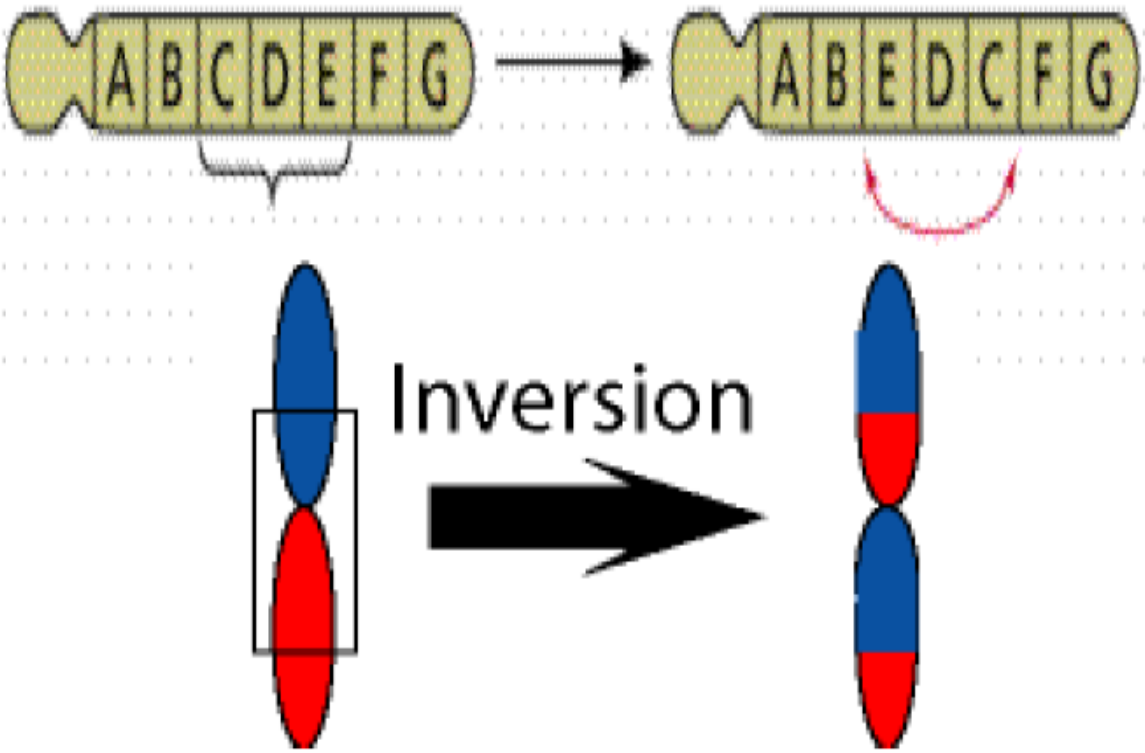
A segment of genes is copied twice and added to the chromosome

Causes:

Charcot–Marie–Tooth disease

(high arched foot, claw feet, confined to a wheelchair)

Chromosomal Inversion



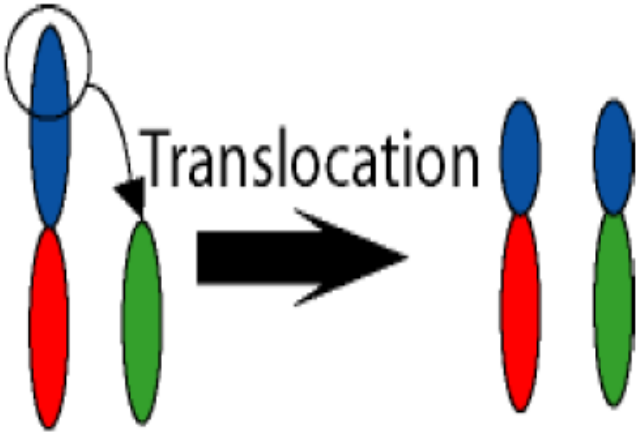
a segment of genes flip end-to-end on the chromosome

Causes:

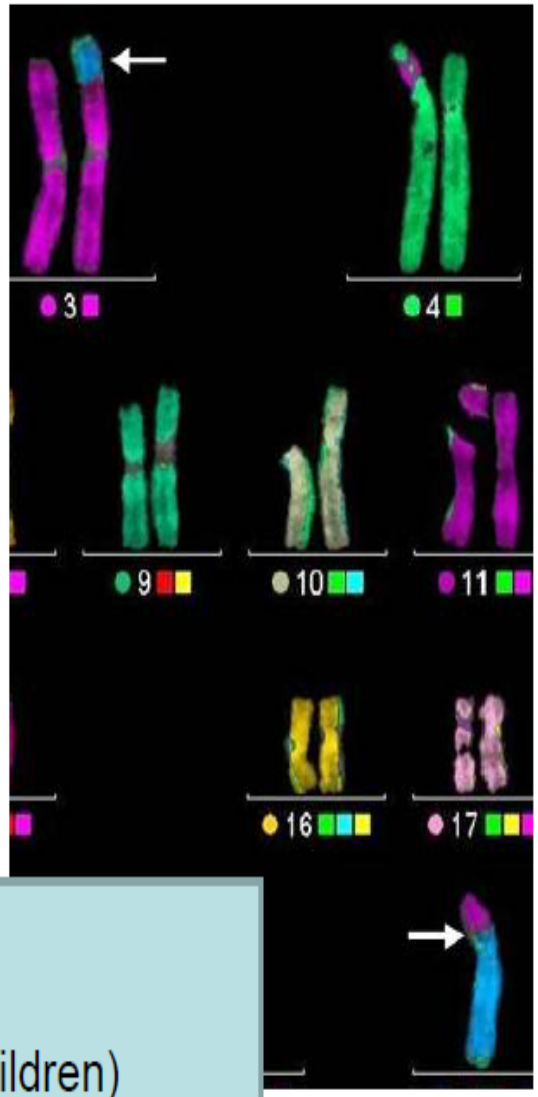
Four-Ring Syndrome

(cleft pallate, club feet, testes don't descend)

Chromosomal Translocation



Material is swapped with another chromosome



Causes:

Burkitt's Lymphoma

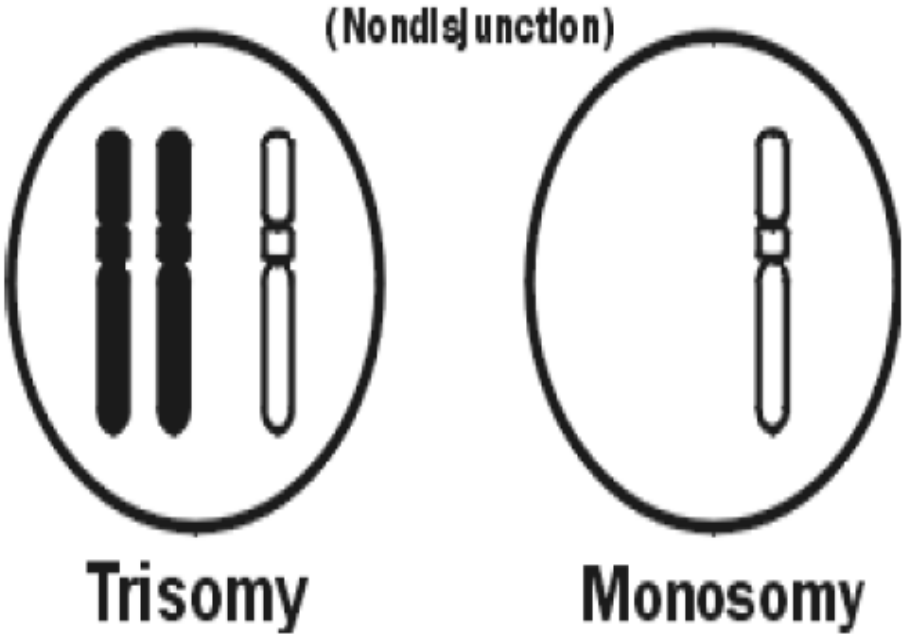
(cancer of the lymph nodes, in children)

- Chromosome numbers can vary in two main ways
  - Euploidy
    - Variation in the number of complete sets of chromosome
  - Aneuploidy
    - Variation in the number of particular chromosomes within a set

  - Euploid variations occur occasionally in animals and frequently in plants
  - Aneuploid variations, on the other hand, are regarded as abnormal conditions

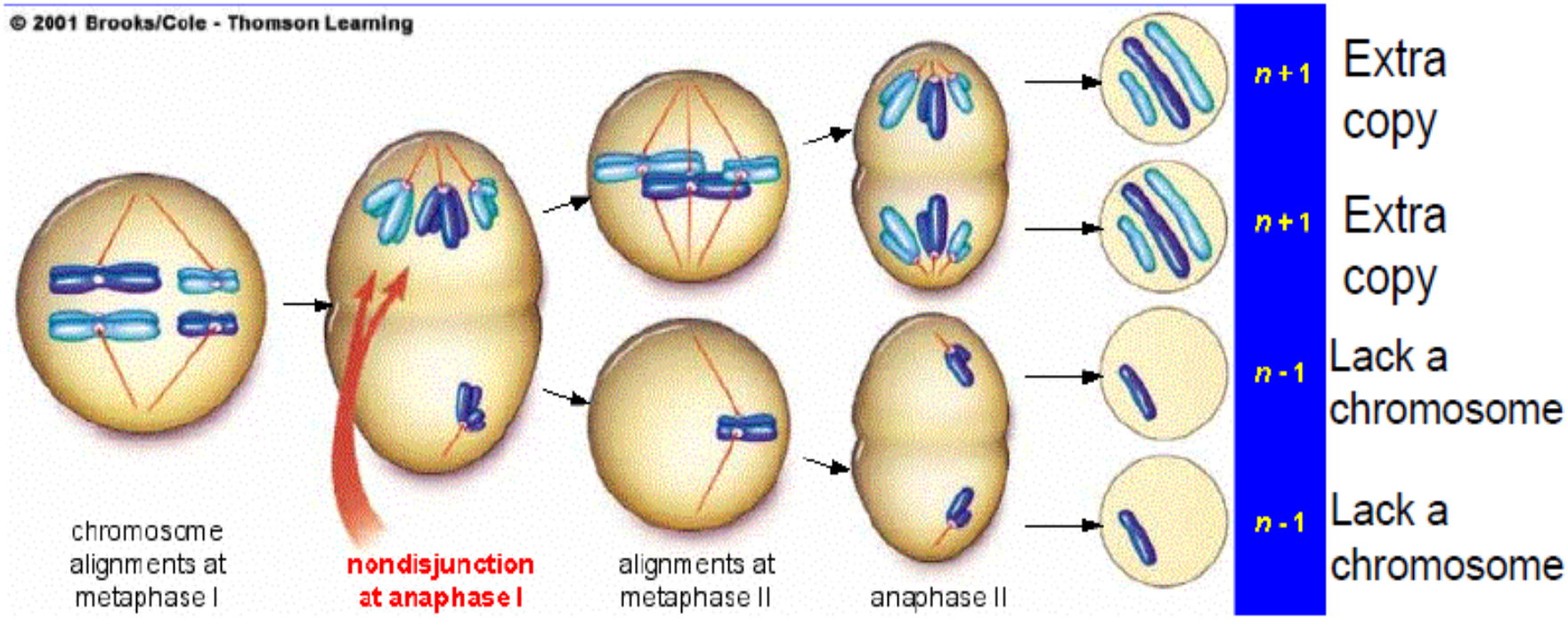
Nondisjunction

- Produces gametes (and therefore a baby) with one missing chromosome **or** one extra chromosome



Nondisjunction: Chromosomes FAIL TO SEPARATE during meiosis

These abnormal gametes are formed when a spindle fibre fails and one of the pair of homologous chromosomes fail to become separated



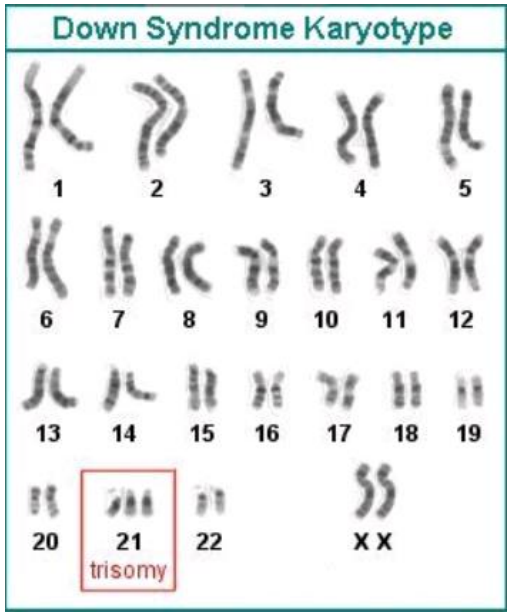
Chromosomal Mutations

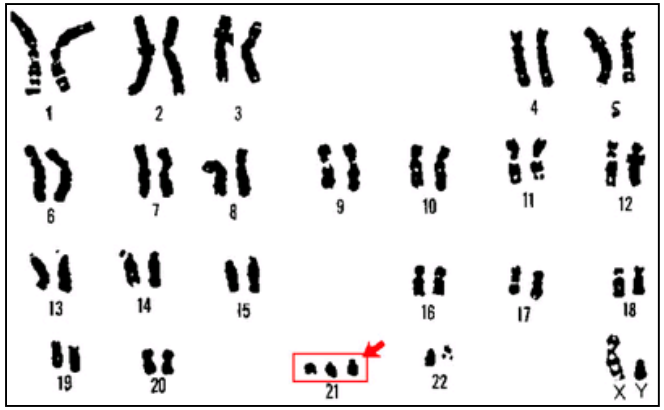
- Most chromosomal mutations are *lethal*
- *If* the fetus survives: Tend to cause wide-spread abnormalities
- *Example:* Down Syndrome

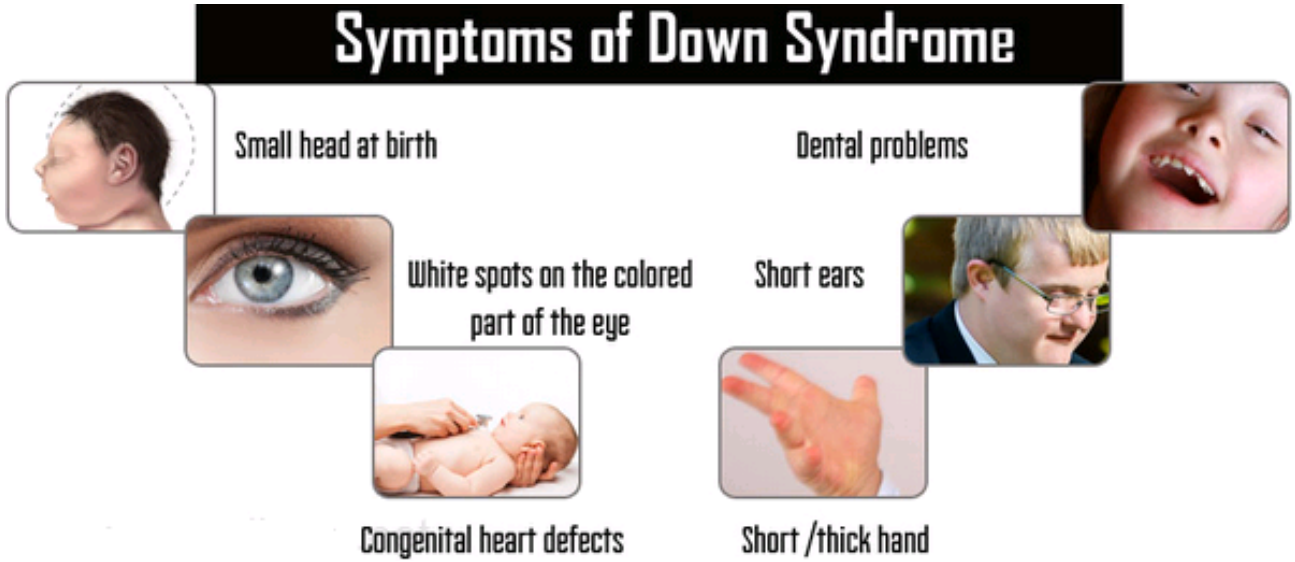
Down Syndrome

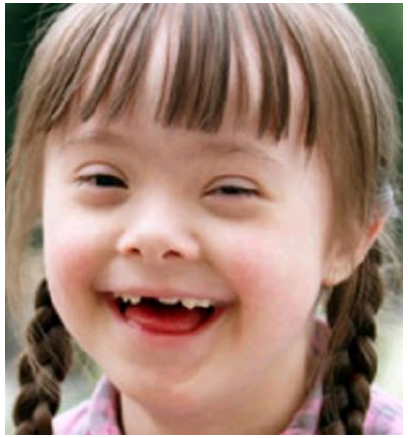
Cause: Nondisjunction of chromosome 21

=> Instead of two copies there are THREE copies of chromosome 21 = TRISOMY 21







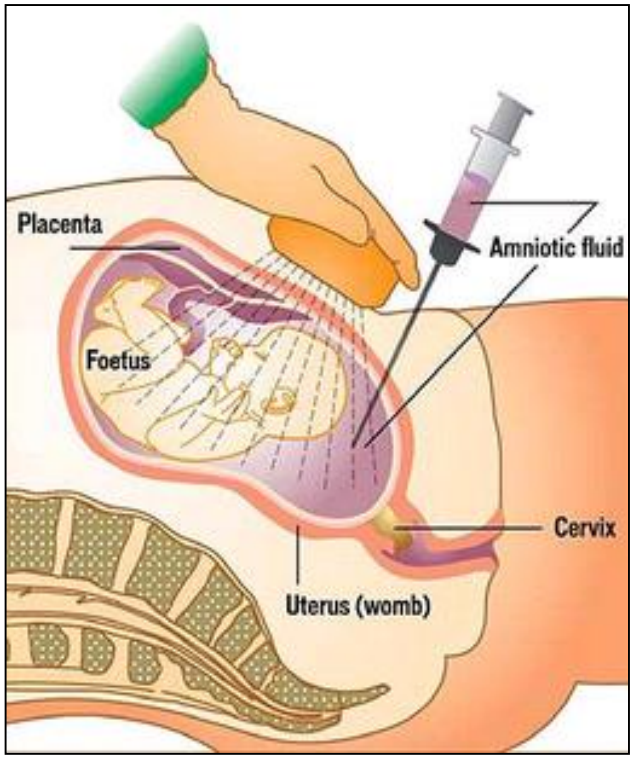


Affected individual is characterized by mental retardation and distinctive physical features

Down Syndrome affects 1 in every 800 babies

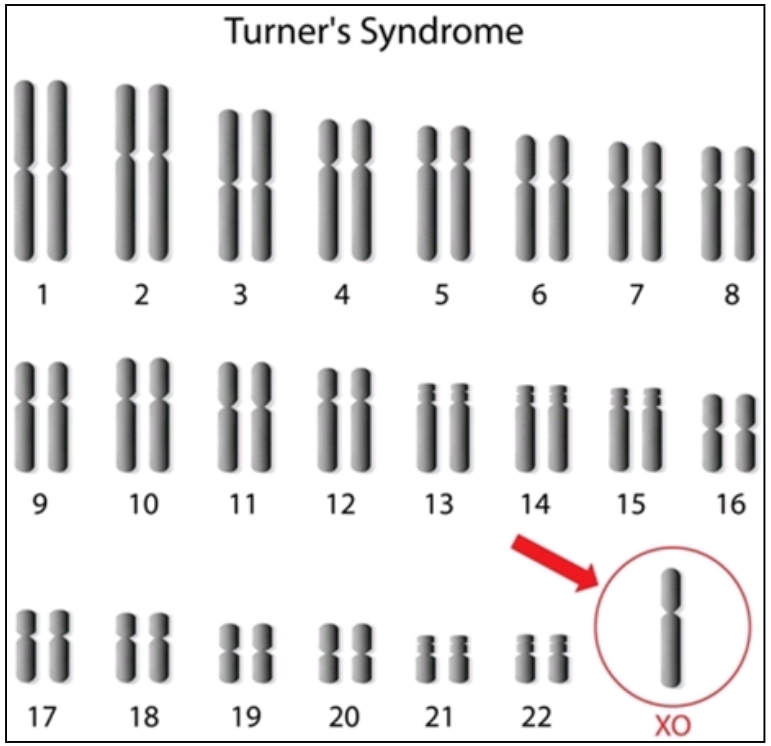
An abnormal egg (n=24) + normal sperm (n=23)
=> Abnormal zygote (n=47)

Egg mother cells of older (≥35) woman (80% due to maternal age tend to be more prone to non-disjunction at meiosis)



Amniocentesis is used to detect possible abnormalities in unborn children.

Turner's Syndrome- Female Only



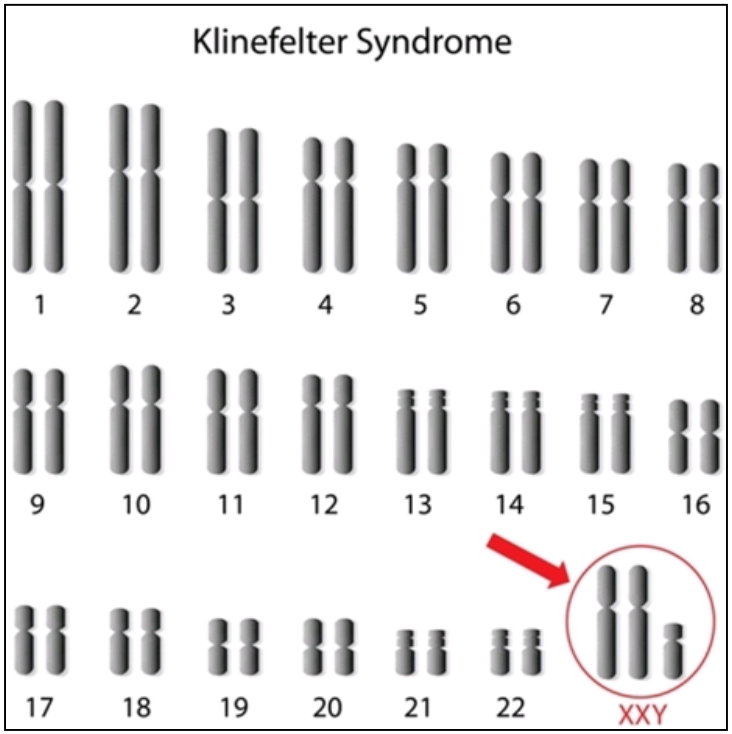
If a gamete which possesses no sex chromosomes meets and fuses with a normal X gamete, the zygote formed has the following chromosome complement - **2n = 44 + XO**

- The reason behind the absence of an X chromosome may be because of an error in the father's sperm or in the mother's egg. This results in every cell having only one X chromosome.
- In certain cases, an error occurs in cell division during initial stages of foetal development. This results in some cells having two copies of the X chromosome. Other cells contain only one copy of the X chromosome.
- Individuals are always female and short in stature

- Their ovaries do not develop so they are infertile and fail to develop secondary sexual characteristics e.g. breast development and menstruation.
- Happens 1:2500 live births

Webbed neck, unusual fingers, short stature, low neckline are all features of the condition. Heart, hearing and visual problems can also occur

Klinefelter's Syndrome- Male Only (47, XXY)



- Happens when:
  - An XX egg is fertilised by a normal Y sperm
  - OR an normal X egg is fertilised by an XY sperm
- **Resulting with chromosome complement 2n = 44 + XXY**
- Individuals are always male and possess male sex organs
- However they are infertile since their testes only develop to half the normal size and fail to produce sperm
- Testes produce low levels of testosterone so facial hair, deepening of voice are only weakly expressed. Some sufferers develop small breasts.
- Occurs in 1:1000 live male births

KEY POINT

Too much or too little DNA is bad!